About This Manual

This manual is for school owners, administrators, teachers, office staff, librarians and hostel wardens. It explains what each module does, how to complete the most common tasks, and how to keep the system secure.

Conventions used

- **Bold** text is a button, field or menu label you see on screen.
- Menu paths read left to right, for example: Sidebar → Student & Parent Details → Student Directory.
- Not every user sees every menu. What you see depends on your role and the permissions assigned to it (see Chapter 5).

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1. Introduction

The School Management System is a web-based School ERP built in PHP 8.2+ with a clean MVC structure, a MySQL or MariaDB database and a Bootstrap 5 interface. It supports multiple schools, each with multiple branches, from a single installation.

What you can do with it

Area	Capabilities
Institution	Schools, branches, holidays, vacations
People	Students, parents, employees, user accounts
Academics	Academic years, classes, sections, subjects, timetable
Daily operations	Student attendance, staff attendance, homework
Assessment	Exam schedules, marks, marksheets, results
Finance	Fees, receipts, accounts and vouchers, payroll
Campus	Transport, inventory, announcements, notifications
Residential and library	Full Hostel and Library modules
Governance	Roles, page and button permissions, audit logs, themes
Output	PDF and Excel exports, certificates, a versioned REST API

2. Installation and Setup

Requirements

Component	Requirement
PHP	8.2 or higher
Database	MySQL 5.7+ or MariaDB 10.4+
Web server	Apache with mod_rewrite enabled (XAMPP or WAMP works for local use)
Libraries	dompdf (PDF) and PhpSpreadsheet (Excel), installed with Composer

Installation steps

1. Copy the project folder into your web server root, for example C:\xampp\htdocs\school-management.
2. Copy the file **.env.example** to **.env**.
3. Open **.env** and check the values below.
4. Start Apache and MySQL, then open the application URL in your browser.

Setting	Default	Meaning
APP_NAME	School Management System	Name shown in the application
APP_ENV	local	Use production on a live server
APP_DEBUG	true	Set to false on a live server
APP_URL	http://localhost/school-management/public	Address of the public folder
DB_HOST / DB_PORT	127.0.0.1 / 3306	Database server
DB_DATABASE	school_management	Database name
DB_USERNAME / DB_PASSWORD	root / (empty)	Database login
TIMEZONE	Asia/Kolkata	Time zone for dates and attendance
SESSION_LIFETIME	28800	Session length in seconds (8 hours)

Automatic database setup

No manual SQL import is needed. On first access the application creates the database if it does not exist, creates any missing tables without touching existing data, and creates the default Owner and Super Admin accounts. If a table is accidentally dropped, it is recreated safely on the next start.

> **Production tip:** set APP_ENV to production and APP_DEBUG to false, and point the web server at the public folder only.

3. Signing In

1. Open the application URL. You land on the **Login** page.
2. Enter your **Username or Email** and **Password**.
3. Select **Sign In**. You are taken to the Dashboard (or to school and branch selection, see below).

Default accounts

Account	Username	Password	Access
System Owner	owner	owner123	Full system access
Super Admin	superadmin	super123	Full administrative privileges

> **Important:** change both default passwords immediately after your first sign-in.

Selecting a school and branch

If your account can access more than one school or branch, you are asked to choose a **school** and then a **branch** after signing in. You can change this at any time with the switch school and switch branch options, and everything you see afterwards is scoped to that choice.

Forgotten password

Select **Forgot password** on the login page, enter your account details, and follow the reset link to choose a new password.

Account lockout

Five failed sign-in attempts lock the account for 15 minutes. The message shows how many minutes remain. There is also a higher limit that protects against repeated attempts from a single network address.

Signing out

Use **Logout** from the profile menu. Sessions also expire automatically after the configured lifetime (8 hours by default).

4. Navigating the Interface

The application has a collapsible **sidebar**, a top **navbar** and a main content area. The sidebar groups pages into categories:

Category	What it contains
College Details	Add College, View College, Branch Manage, Holiday Assign
Student & Parent Details	Student Directory, Parent Directory
Class & Subjects	Classes & Sections, Subjects Directory
Academics	Timetable Generator, Academic Years, Attendance Portal
Homework Portal	Dashboard, All Homework, Create, Drafts, Published, Due, Overdue, Submissions, Evaluation, Reports
Examination	Exam Time Table, Enter Exam Marks
Finance	Fees Collection, Financial Accounts, Employee Payroll
People	Employees & Staff, Shift Management
Staff Attendance	Self punch, mark and manage, bulk sheet, daily, dashboard, monthly, regularization, history, shifts, leaves, reports, settings
Campus Services	Transport Routes, Inventory Assets, Announcements
Document/Photos	Document Category, Document Assign
Reports	Reports Hub, Certificates
User Account	User List, Role List, Role & Permission
System	Settings, Theme & Appearance, Audit Logs
Hostel Management	Navbar module with its own dashboard and pages
Library Management	Navbar module with its own dashboard and pages

Notifications. The bell icon shows an unread count. Open it to read, mark one as read, dismiss, or **mark all read**.

Profile. Open **Profile** to update your own details and change your password.

Tables. Most list pages offer filters (for example class, section, gender, status), search, row action buttons (view, edit, toggle status, delete) and export buttons. Destructive actions ask for confirmation first.

5. Roles and Permissions

Access is controlled by roles. A role bundles permissions; each user has a role. Permissions are fine-grained: they control which **pages** appear and which **header buttons, row action buttons and filters** are usable on each page.

Built-in roles

Role	Purpose
owner	System owner with full access (all_access permission)
superadmin	Full administrative privileges (all_access permission)
student	Created automatically for student portal accounts
parent	Created automatically for parent portal accounts
teacher	Used for teaching staff accounts

You can create further custom roles, such as accountant, librarian or warden, from **Role List**.

Creating a role

1. Go to Sidebar → User Account → **Role List**.
2. Select **Add Role**, enter a **name** and **description**, and save.

Assigning permissions

1. Go to Sidebar → User Account → **Role & Permission**.
2. Choose the role. Permissions are shown as cards that follow the sidebar categories.
3. Tick the pages the role may open, then the buttons it may use: Add, View, Edit, Delete and Print style actions are listed per page.
4. Save.

You can also give a **single user** their own extra permissions, and manage additional assignments for that user, without changing the role for everyone else. A page access statistics view shows which pages are being used.

> **Best practice:** give each role only what it needs. For example, a librarian needs the Library pages and not Finance.

6. College, Branches and Holidays

Schools (College Details)

- **Add College** registers a new school or college, with its credentials and optional branches.
- **View College** lists all schools. Row actions: view profile, edit, activate or deactivate, delete.

Branches

- **Branch Manage** lists campuses. Add a branch, view contact information, edit, activate or deactivate, or delete.

Holidays and vacations

1. Open **Holiday Assign**.
2. **Add Holiday** to create a single entry, or **Adopt Public Holidays** to take gazetted public holidays in one step.
3. **Create Vacation** schedules a longer break. The system calculates the affected days and checks for conflicts with exams before saving.
4. Holidays can apply to selected classes and sections. Use the toggle to approve a day as a holiday or keep it a working day.

Holidays feed into attendance and exam scheduling, so set them up before building exam timetables.

7. Students and Parents

Admitting a student

1. Go to Sidebar → Student & Parent Details → **Student Directory**.
2. Select **Admit New Student**.
3. Complete the form sections shown in the table below.
4. Save. The student appears in the directory.

Section	Typical fields
Admission	Academic Year, Class, Section, Admission Number, GR Number, Admission Date
Personal	First Name, Last Name, Gender, Date of Birth, Blood Group, Caste / Category
Contact	Mobile Number, Alternate Mobile, Email, Current and Permanent Address, City, Country
Father, Mother, Guardian	Name, education, occupation, annual income, photo; guardian relation
Emergency	Contact person, relation, phone
Parent account	Link an existing parent (Search Parent) or create a parent username, email and password
Student login	Optional username and password for the student portal
Documents	Required documents with expiry dates, based on your Document Assign rules

Working with the directory

- **Filters:** class, section, gender and status.
- **Row actions:** quick preview, full profile, edit, activate or deactivate, delete.
- **Export Roster** prints or exports the filtered list.

Parents

Parent Directory lists guardians and their linked students. New parents are normally created during student admission, and you can edit or remove them afterwards.

8. Academics: Years, Classes, Subjects, Timetable

Set these up in order, because later steps depend on earlier ones.

1. **Academic Years.** Add the session and its term periods.
2. **Classes & Sections.** Add classes (grades), then sections for each class and assign a class teacher. You can change a section teacher later.
3. **Subjects Directory.** Add subjects with codes, assign them to classes (singly or in bulk), and assign teachers to subjects.
4. **Timetable Generator.** Build the weekly schedule: choose class and section, place subjects and teachers into periods, and save. Use view mode to review, and clear or delete to rebuild.

9. Student Attendance

Open Sidebar → Academics → **Attendance Portal**.

1. Choose the **class**, **section** and **date**.
2. The student list loads. Mark each student as **Present**, **Absent**, **Late** or **Half Day**.
3. Save.

Use **View** to review a day and **Monthly** for a month grid with counts of present, absent, late and half-day. Both can be exported as PDF or Excel.

10. Homework Portal

A complete workflow from setting work to grading it.

Stage	Page	What happens
Compose	Create Homework	Select class, section and subject, add description and files
Draft	Drafts	Unpublished work you are still preparing

Stage	Page	What happens
Publish	Published	Open to students; you can unpublish, close or archive
Track	Due / Pending, Overdue	Approaching deadlines and late work
Collect	Submissions	Student turn-ins and uploaded documents
Grade	Evaluation	Teacher grading queue with feedback notes
Analyse	Dashboard, Reports	Completion rates and class compliance, printable

For each student you can mark work **completed**, **incomplete** or **excused**, or **reset** the status. Bulk evaluate and bulk status updates save time for large classes.

11. Examinations

Exam timetable

1. Open Sidebar → Examination → **Exam Time Table**.
2. Create an exam: select class and subjects, then set dates, times and rooms.
3. Use **conflict checks** to catch clashes and holiday overlaps. **Smart generate** can propose a schedule automatically.
4. Edit or delete a timetable at any time.

Postponement

You can postpone a single paper or a whole batch. A preview shows the effect before you confirm, and **postponement history** keeps a record.

Marks and results

Open **Enter Exam Marks**, select the exam, class and section, and enter scores. Related pages:

- **Marksheet:** printable report card per student.
- **Progress:** a student's performance across exams.
- **Results:** class results with grades.
- **Reports:** summary analytics.

12. Finance: Fees, Accounts and Payroll

Fee setup

Open **Fees Collection** and add **fee categories** (such as tuition) and **fee structures** that define what each class pays.

Collecting a fee

1. Go to Finance → Fees Collection → **Payments**.

2. Select the **Student**.
3. Enter **Amount Paid** and **Payment Date**.
4. Choose the **Payment Mode**: Cash, Cheque, Online, UPI, Card or Bank transfer.
5. For non-cash modes add the **Transaction Reference / Cheque No.**
6. Save, then open or print the **receipt**.

The page also shows dues status for each student.

Financial Accounts

Maintain the chart of accounts with **account heads**, and record debit and credit entries as **vouchers**. This gives the institution a ledger of its income and expenses.

Employee Payroll

Employee Payroll calculates monthly salaries with components and deductions. Choose the month, **generate** payroll, and review payslips.

13. Staff, Shifts and Staff Attendance

Adding an employee

Go to People → **Employees & Staff** → Add. The form covers:

Section	Fields
Employment	Employee Code, Staff Category / Type, Department, Designation, Role, Joining Date, contract dates
Personal	Name, Gender, Date of Birth, Blood Group, Nationality, Phone, Email, Addresses
Professional	Highest Qualification, Experience (Years), Specialization
Emergency	Contact name and phone
Bank	Account holder, bank name, account number, IFSC, branch address, account type, UPI ID
Photo and documents	Profile photo and required documents with expiry dates
Login	Tick Create Account Credentials for this Employee? to create a username and password

Shift management

Define shifts, then assign them to a category of staff or to one employee. Bulk assignment, shift history and upcoming shifts are available, and an employee's regular shift can be restored after a temporary change.

Staff attendance

Task	How
Self punch	Staff check in and out themselves; the system can validate GPS location and IP address
Mark attendance	Supervisors mark individuals or whole departments
Regularization	Staff request a correction for a missed punch; supervisors approve
Monthly view	Calendar of punches and holidays
Reports	Monthly ledgers, overtime and shortfall
Settings	Geofence radius, IP restrictions and punch policies

A correction history keeps a record of every change.

14. Campus Services

- **Transport Routes.** Add vehicles and routes, then assign students to a route and seat.
- **Inventory Assets.** Add items with quantities, then assign items to people or places and track issuances.
- **Announcements.** Compose a notice and send it to selected audiences using saved templates; recipients also receive an in-app notification.

15. Documents, Certificates and Reports

Required documents

1. **Document Category:** create the types of file you collect (for example ID proof).
2. **Document Assign:** decide which roles or staff tiers must submit each category. Assignments can be activated or deactivated per role.

The student and employee forms then ask for exactly those documents.

Certificates

Use **Certificates** to generate transfer certificates, bonafide letters and achievement diplomas. Select the template and student, generate, then open and print the result.

Reports Hub

The **Reports Hub** is the analytics centre for academic, fee and student metrics, with printable summaries.

16. Hostel Management

Open the **Hostel** module from the top navigation.

Page	Purpose
Hostel Dashboard	Occupancy and key statistics
Hostels & Blocks	Buildings, wings and floors
Rooms & Beds	Capacity, bed numbers, AC or Non-AC, maintenance status
Bed Allocations	Assign a student to a bed; transfer or check out; view room history
Daily Roll Call	Evening attendance
Leaves & Outpass	Request, approve or reject night and weekend leave; record return
Hostel Fees & Billing	Generate monthly invoices, collect fees, issue receipts
Gate Visitors	Log visitors and record checkout time
Complaints & Maintenance	Log issues, assign staff, mark resolved
Hostel Reports	Occupancy, vacancies, fee recovery, attendance

Typical flow: create a block, add rooms and beds, allocate a student, generate the monthly fee, and collect payment.

17. Library Management

Open the **Library** module from the top navigation.

Page	Purpose
Library Dashboard	Circulation trends and catalogue size
Book Inventory	Titles, authors, ISBN, publisher, shelf location
Book Copies	Individual copies with accession numbers, barcodes and condition
Circulation Desk	Issue, return and renew books
Library Members	Student and staff cards, current borrowings, limits
Fines Management	Collect or waive overdue fines
Library Setup & Categories	Categories and loan rules
Library Reports	Circulation, overdue lists, losses, popular titles

Loan rules (Library Settings)

Setting	Meaning
library_issue_days	Standard loan period in days
library_max_renewals	Renewals allowed per loan
library_max_student_books	Books a student can hold
library_max_staff_books	Books a staff member can hold

Setting	Meaning
library_fine_per_day	Fine charged per overdue day

Issuing and returning

1. Open **Circulation Desk** → Issue.
2. Pick the member and the book copy, then confirm.
3. On return, open the loan and select **Return**. Overdue items create a fine that you collect from **Fines Management**.

18. Users, Settings and Appearance

User accounts

User List shows every login. You can add a user, edit, activate or deactivate, or delete. Student, parent and employee accounts can also be created directly from their own forms, or later from User List by choosing the person and generating credentials. The system checks that a username is free as you type.

Settings

Settings holds school branding (logo), contact details, currency and date formats. Further tabs cover **branding**, **layout** and **navigation** options.

Theme & Appearance

Choose a ready-made preset, or customize colors, typography, sidebar, navbar and components.

Preset	Style
Pearl Luxury	Light, refined default
Champagne Gold	Warm gold accents
Royal Navy	Deep blue
Emerald Executive	Green professional
Obsidian Dark	Dark mode
Crystal Glass	Translucent glass look

Use **Apply** to activate a preset, **Customize** to fine-tune it, **Reset** to return to defaults, and **Export** to save the theme.

19. Exporting Data

Most list and report pages offer **PDF** and **Excel** export. PDFs use your institution name and logo and respect the page orientation set for that report. Exports are permission-controlled, so a user can only export data they are allowed to see.

Exportable areas include classes, sections, subjects, timetables, students, parents, employees, student and staff attendance, shifts, payroll, exams and marksheets, fees and payments, accounts, homework, hostel, library, transport, inventory, schools, branches, holidays, users, roles, documents, audit logs and the Reports Hub.

20. Security and Audit Trail

Protection	How it works
Sessions	Server-side sessions with automatic expiry
CSRF	Every state-changing request carries an anti-CSRF token
SQL safety	All queries use prepared statements
Uploads	Extension blacklist, MIME check, size limit, random filenames, isolated folder
Brute force	5 failed attempts lock an account for 15 minutes
Error handling	Users see a friendly error; technical details go to storage/logs/app.log
Safe start-up	Never drops tables or data during normal use

Audit Logs (System → Audit Logs) record sign-ins, creations, edits and deletions. A security log also records events such as account lockouts. Review them regularly.

21. Troubleshooting and FAQ

Problem	What to try
Blank page or 503 error	Check that MySQL is running and the .env database values are correct; see storage/logs/app.log
Links return 404	Enable Apache mod_rewrite and make sure .htaccess is allowed in the public folder
Cannot sign in	Check username or email and password; wait 15 minutes if the account is locked
A menu item is missing	Your role lacks that permission; ask an administrator to update Role & Permission
A button is missing on a page	Button-level permission is off for your role
Dates or times look wrong	Set TIMEZONE in .env and restart Apache

Problem	What to try
PDF or Excel export fails	Confirm the vendor folder is present; run composer install if it is not
Upload rejected	The file type or size is not allowed; use a common format such as PDF, JPG or PNG
Session ends too soon	Increase SESSION_LIFETIME in .env

Can I run several schools in one installation? Yes. Add each school and its branches, then select the one to work in after sign-in.

Will restarting wipe my data? No. Start-up only creates what is missing.

22. Appendix

A. Recommended setup order

1. Change default passwords and set .env values.
2. Add the school, branches and branding.
3. Add holidays and the academic year.
4. Add classes, sections, subjects and the timetable.
5. Create document categories and assignments.
6. Create roles and permissions.
7. Add employees, then students and parents.
8. Set up fees; start attendance, homework and exams.
9. Enable Hostel and Library if you use them.

B. REST API (v1)

Base path: **/api/v1/** through the front controller public/api.php. Authentication uses the PHP session cookie. Requests that change data must send the CSRF token in the X-CSRF-TOKEN header or a csrf_token field. Use application/json, or multipart/form-data with method spoofing for file uploads.

Endpoint	Purpose
POST /api/v1/auth/login	Sign in with username and password
POST /api/v1/auth/logout	Sign out
GET /api/v1/auth/me	Current user profile

Resources such as students, employees, attendance, homework, hostel, library, themes, roles and users follow the same pattern. Every response uses one structure with the fields **success**, **message**, **data** and **errors**. Status codes: 200 and 201 for success, 422 for validation errors, and 400, 401, 403, 404 and 500 for failures. See docs/api.md in the project for the full list.

C. Support checklist

When asking for help, note your role, the page you were on, what you expected, what happened, and the time. The audit log and storage/logs/app.log help administrators find the cause quickly.